

Seminar 1. Bayesian Reasoning: Solutions

Bayesian Methods

1. A medical test finds a serious disease in a patient. The test is 99% accurate: if the patient has the disease, the test is positive with probability 99%; if the patient does not have the disease, the test is negative with probability 99%. The disease is rare: only one person in 10 000 has it. Find the probability that the patient really has the disease.

Interpretation. $D = 1$: the patient has the disease. $T = 1$: the test result is positive.

$$p(D = 1) = 10^{-4}, \quad p(T = 1 \mid D = 1) = 0.99, \quad p(T = 0 \mid D = 0) = 0.99.$$

Find $p(D = 1 \mid T = 1)$.

Idea. The disease causes the test result: the graph is $D \rightarrow T$. The source node and the edge give the joint distribution:

$$p(D, T) = p(D) p(T \mid D).$$

The joint distribution gives any other probability with two operations:

$$\begin{aligned} p(T) &= \sum_D p(D, T) && \text{(marginal; for continuous } x: p(y) = \int p(x, y) dx), \\ p(D \mid T) &= \frac{p(D, T)}{p(T)} && \text{(conditional).} \end{aligned}$$

Together, they give Bayes' theorem:

$$p(D \mid T) = \frac{p(D, T)}{p(T)} = \frac{p(D, T)}{\sum_{D'} p(D', T)} = \frac{p(D) p(T \mid D)}{\sum_{D'} p(D') p(T \mid D')}.$$

Solution.

Step 1. Write the goal.

$$p(D = 1 \mid T = 1) = \frac{p(D = 1, T = 1)}{p(T = 1)}.$$

Step 2. Find the numerator $p(D = 1, T = 1)$.

$$p(D = 1, T = 1) = p(D = 1) p(T = 1 \mid D = 1) = 0.0001 \cdot 0.99 = 0.000099.$$

Step 3. Find the denominator $p(T = 1)$.

$$p(T = 1) = p(D = 1, T = 1) + p(D = 0, T = 1) = 0.000099 + p(D = 0, T = 1) \quad (\text{Step 2}).$$

Step 3a. Find $p(D = 0, T = 1)$ for Step 3.

$$\begin{aligned} p(D = 0, T = 1) &= p(D = 0) p(T = 1 \mid D = 0), \\ p(D = 0) &= 1 - p(D = 1) = 1 - 0.0001 = 0.9999, \\ p(T = 1 \mid D = 0) &= 1 - p(T = 0 \mid D = 0) = 1 - 0.99 = 0.01 && \text{(false positive),} \\ p(D = 0, T = 1) &= 0.9999 \cdot 0.01 = 0.009999. \end{aligned}$$

Step 3b. Substitute Step 3a into Step 3.

$$p(T = 1) = 0.000099 + 0.009999 = 0.010098.$$

Step 4. Substitute Steps 2 and 3 into Step 1.

$$p(D = 1 \mid T = 1) = \frac{0.000099}{0.010098} = \frac{99}{99 \cdot 102} = \frac{1}{102} \approx 0.0098.$$

Conclusion. The probability increases from 0.01% to only 1%. In 10 000 people, the test finds $10\,000 \cdot 0.000099 \approx 1$ sick person and $10\,000 \cdot 0.009999 \approx 100$ healthy people. Thus most positive results of a mass test for a rare disease are false. A second positive independent test uses $1/102$ as the prior:

$$p(D = 1 \mid T_1 = 1, T_2 = 1) = \frac{\frac{1}{102} \cdot 0.99}{\frac{1}{102} \cdot 0.99 + \frac{101}{102} \cdot 0.01} = \frac{0.99}{0.99 + 1.01} = 0.495.$$

2. Consider the following probabilistic model. At a seminar, a student answers badly ($A = 1$) or well ($A = 0$). This depends on whether the student has depression ($D = 1$ or $D = 0$) and whether the student went to a party the night before ($P = 1$ or $P = 0$). The party can also give the student a headache ($H = 1$). A bad answer makes the teacher unhappy ($T = 1$). Figure 1 shows the causal relations in the model. The probabilities are:

$p(A = 1 \mid P, D) \mid P \quad D$	$p(H = 1 \mid P) \mid P$	$p(T = 1 \mid A) \mid A$
0.999	0.9	0.95
0.9	0.2	0.5
0.9		
0.01		

$$p(P = 1) = 0.2, \quad p(D = 1) = 0.4.$$

Find $p(P = 1 \mid H = 1)$, $p(P = 1 \mid T = 1)$, and $p(P = 1 \mid T = 1, H = 1)$.

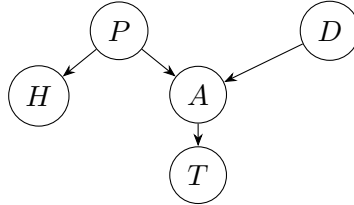


Figure 1: Causal relations for Problem 2

Idea. Use the method of Problem 1 on a larger graph.

Joint distribution. Apply the chain rule one variable at a time, from the children to the parents. A missing arrow means no direct influence: given its parents, a node does not depend on the other nodes in the condition.

$$\begin{aligned}
 p(P, D, A, H, T) &= p(P, D, A, H) p(T \mid P, D, A, H) \\
 &= p(P, D, A, H) p(T \mid A) && \text{(parent of } T: A) \\
 p(P, D, A, H) &= p(P, D, A) p(H \mid P, D, A) \\
 &= p(P, D, A) p(H \mid P) && \text{(parent of } H: P) \\
 p(P, D, A) &= p(P, D) p(A \mid P, D) && \text{(parents of } A: P, D) \\
 p(P, D) &= p(P) p(D \mid P) \\
 &= p(P) p(D) && (D \text{ has no parents})
 \end{aligned}$$

Thus

$$p(P, D, A, H, T) = p(P)p(D)p(A | P, D)p(H | P)p(T | A).$$

Each factor is a given table.

Queries. Move along the edges with sums, and against the edges with Bayes' theorem:

$$p(T | P) = \sum_A p(T | A)p(A | P), \quad p(P | T) = \frac{p(P)p(T | P)}{\sum_{P'} p(P')p(T | P')}.$$

A sum over a node without children removes its factor, because $\sum_x p(x | \text{parents}) = 1$.

Solution.

Step 1. Find $p(P = 1 | H = 1)$.

$$p(P = 1 | H = 1) = \frac{p(P = 1, H = 1)}{p(H = 1)} = \frac{p(P = 1, H = 1)}{p(P = 1, H = 1) + p(P = 0, H = 1)}.$$

Step 1a. Find $p(P, H = 1)$ for *Step 1*. Sum out T, A, D (each factor sums to 1):

$$\begin{aligned} p(P, H) &= p(P)p(H | P), \\ p(P = 1, H = 1) &= 0.2 \cdot 0.9 = 0.18, \\ p(P = 0, H = 1) &= (1 - 0.2) \cdot 0.2 = 0.8 \cdot 0.2 = 0.16. \end{aligned}$$

Step 1b. Substitute *Step 1a* into *Step 1*.

$$p(P = 1 | H = 1) = \frac{0.18}{0.18 + 0.16} = \frac{0.18}{0.34} = \frac{9}{17} \approx 0.529.$$

Step 2. Find $p(P = 1 | T = 1)$.

$$p(P = 1 | T = 1) = \frac{p(P = 1, T = 1)}{p(P = 1, T = 1) + p(P = 0, T = 1)}, \quad p(P, T) = p(P)p(T | P).$$

Step 2a. Find $p(T = 1 | P)$ for *Step 2*.

$$\begin{aligned} p(T | P) &= \sum_A p(T, A | P) = \sum_A p(T | A, P)p(A | P) \\ &= \sum_A p(T | A)p(A | P) \end{aligned} \quad (\text{parent of } T: A).$$

Step 2b. Find $p(A = 1 | P)$ for *Step 2a*.

$$\begin{aligned} p(A | P) &= \sum_D p(A, D | P) = \sum_D p(A | P, D)p(D | P) \\ &= \sum_D p(A | P, D)p(D) \end{aligned} \quad (D \text{ has no parents}).$$

$$\begin{aligned} p(A = 1 | P = 1) &= 0.999 \cdot 0.4 + 0.9 \cdot (1 - 0.4) = 0.3996 + 0.54 = 0.9396, \\ p(A = 1 | P = 0) &= 0.9 \cdot 0.4 + 0.01 \cdot (1 - 0.4) = 0.36 + 0.006 = 0.366. \end{aligned}$$

Step 2c. Substitute *Step 2b* into *Step 2a*.

$$\begin{aligned} p(T = 1 | P = 1) &= 0.95 \cdot 0.9396 + 0.5 \cdot (1 - 0.9396) = 0.89262 + 0.0302 = 0.92282, \\ p(T = 1 | P = 0) &= 0.95 \cdot 0.366 + 0.5 \cdot (1 - 0.366) = 0.3477 + 0.317 = 0.6647. \end{aligned}$$

Step 2d. Substitute Step 2c into Step 2.

$$\begin{aligned} p(P = 1, T = 1) &= 0.2 \cdot 0.92282 = 0.184564, \\ p(P = 0, T = 1) &= 0.8 \cdot 0.6647 = 0.53176, \\ p(P = 1 \mid T = 1) &= \frac{0.184564}{0.184564 + 0.53176} = \frac{0.184564}{0.716324} \approx 0.258. \end{aligned}$$

Step 3. Find $p(P = 1 \mid T = 1, H = 1)$.

$$p(P = 1 \mid T = 1, H = 1) = \frac{p(P = 1, H = 1, T = 1)}{p(P = 1, H = 1, T = 1) + p(P = 0, H = 1, T = 1)}.$$

Step 3a. Find $p(P, H, T)$ for Step 3.

$$\begin{aligned} p(P, H, T) &= \sum_A \sum_D p(P) p(D) p(A \mid P, D) p(H \mid P) p(T \mid A) \\ &= p(P) p(H \mid P) \sum_A p(T \mid A) \sum_D p(D) p(A \mid P, D) \quad (\text{move factors out of the sums}) \\ &= p(P) p(H \mid P) \sum_A p(T \mid A) p(A \mid P) \quad (\text{Step 2b}) \\ &= p(P) p(H \mid P) p(T \mid P) \quad (\text{Step 2a}). \end{aligned}$$

With the values from Step 2c:

$$\begin{aligned} p(P = 1, H = 1, T = 1) &= 0.2 \cdot 0.9 \cdot 0.92282 = 0.166108, \\ p(P = 0, H = 1, T = 1) &= 0.8 \cdot 0.2 \cdot 0.6647 = 0.106352. \end{aligned}$$

Step 3b. Substitute Step 3a into Step 3.

$$p(P = 1 \mid T = 1, H = 1) = \frac{0.166108}{0.166108 + 0.106352} = \frac{0.166108}{0.27246} \approx 0.610.$$

Conclusion.

$$\underbrace{0.2}_{\text{prior}} \rightarrow \underbrace{0.529}_{H=1}, \quad \underbrace{0.258}_{T=1}, \quad \underbrace{0.610}_{H=1, T=1}.$$

An unhappy teacher is weak evidence: $p(T = 1 \mid P = 0) = 0.6647$ is also high. Two observations together give the strongest evidence.

3. Let $x_1 \sim \text{Poiss}(\lambda_1)$ and $x_2 \sim \text{Poiss}(\lambda_2)$ be two independent Poisson random variables, that is, $p(x_i = k) = \exp(-\lambda_i) \frac{\lambda_i^k}{k!}$, $k = 0, 1, 2, \dots$. Prove that $x_1 + x_2 \sim \text{Poiss}(\lambda_1 + \lambda_2)$.

Idea. For $y = f(x_1, \dots, x_n)$ with discrete x_i , split the event $\{y = k\}$ into disjoint events and sum:

$$p(y = k) = \sum_{f(j_1, \dots, j_n) = k} p(x_1 = j_1, \dots, x_n = j_n) = \sum_{f(j_1, \dots, j_n) = k} \prod_i p(x_i = j_i) \quad (\text{if independent}).$$

Then simplify the sum and compare it with known distributions.

Solution. Goal: show that $p(x_1 + x_2 = k) = e^{-(\lambda_1 + \lambda_2)}(\lambda_1 + \lambda_2)^k / k!$ for all $k \geq 0$.

$$\begin{aligned}
p(x_1 + x_2 = k) &= \sum_{j=0}^k p(x_1 = j, x_2 = k - j) && \text{(disjoint events, } x_i \geq 0) \\
&= \sum_{j=0}^k p(x_1 = j) p(x_2 = k - j) && \text{(independence)} \\
&= \sum_{j=0}^k e^{-\lambda_1} \frac{\lambda_1^j}{j!} \cdot e^{-\lambda_2} \frac{\lambda_2^{k-j}}{(k-j)!} && \text{(Poisson probabilities)} \\
&= e^{-(\lambda_1 + \lambda_2)} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!} && \text{(factors without } j \text{ out of the sum)} \\
&= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \frac{k!}{j! (k-j)!} \lambda_1^j \lambda_2^{k-j} && \text{(multiply and divide by } k!) \\
&= \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \binom{k}{j} \lambda_1^j \lambda_2^{k-j} \\
&= e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!} && \text{(binomial theorem).}
\end{aligned}$$

This is $\text{Poiss}(\lambda_1 + \lambda_2)$.

Conclusion. A sum of independent Poisson variables is a Poisson variable, and the parameters add. By induction, this is true for any finite number of variables.

4. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. Find the maximum likelihood estimate of λ .

Idea.

$$L(\lambda) = \prod_i p(x_i | \lambda) \rightarrow \ell(\lambda) = \log L(\lambda) = \sum_i \log p(x_i | \lambda) \rightarrow \ell'(\lambda) = 0 \rightarrow \ell''(\lambda) < 0.$$

The logarithm is increasing, so L and ℓ have the same maximum point.

Solution. Let $S = \sum_{i=1}^N x_i$ and $\bar{x} = S/N$.

Step 1. Write the goal.

$$\hat{\lambda}_{\text{ML}} = \arg \max_{\lambda} L(\lambda) = \arg \max_{\lambda} \ell(\lambda).$$

Step 2. Write $\ell(\lambda)$.

$$\ell(\lambda) = \log \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} = \sum_{i=1}^N (-\lambda + x_i \log \lambda - \log x_i!) = -N\lambda + S \log \lambda - \sum_{i=1}^N \log x_i!.$$

Step 3. Set the derivative to zero.

$$\ell'(\lambda) = -N + \frac{S}{\lambda} = 0 \implies \lambda = \frac{S}{N} = \bar{x}.$$

Step 4. Make sure that it is a maximum.

$$\ell''(\lambda) = -\frac{S}{\lambda^2} < 0 \quad \text{if } S > 0.$$

Thus ℓ is strictly concave, and $\bar{x}$ is the global maximum. If $S = 0$, then $\ell(\lambda) = -N\lambda$ decreases, and the maximum is at $\lambda = 0 = \bar{x}$.

$$\hat{\lambda}_{\text{ML}} = \bar{x}.$$

5. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. The prior distribution is the gamma distribution

$$p(\lambda) = \mathcal{G}(\lambda \mid a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda), \quad a, b > 0.$$

Find the Bayesian estimate of λ as the mean of the posterior distribution $p(\lambda \mid x_1, \dots, x_N)$.

Idea. The numerator and the denominator of Bayes' theorem contain the same function:

$$p(\lambda \mid X) = \frac{p(X \mid \lambda) p(\lambda)}{\int p(X \mid \lambda) p(\lambda) d\lambda}, \quad X = (x_1, \dots, x_N).$$

Do not calculate the integral directly:

$$\begin{aligned} p(X \mid \lambda) p(\lambda) &= C \cdot g(\lambda) && (C \text{ does not depend on } \lambda) \\ g(\lambda) &= Z q(\lambda) && (q \text{ is a known density}) \\ \int CZ q(\lambda) d\lambda &= CZ && (\text{a density integrates to } 1) \\ p(\lambda \mid X) &= \frac{CZ q(\lambda)}{CZ} = q(\lambda). \end{aligned}$$

The same method gives the posterior mean.

Solution. Let $S = \sum_{i=1}^N x_i$, $a' = a + S$, and $b' = b + N$.

Step 1. Write the goal.

$$\hat{\lambda}_B = \mathbb{E}[\lambda \mid X] = \int \lambda p(\lambda \mid X) d\lambda.$$

Step 2. Find $p(\lambda \mid X)$ for Step 1.

$$p(\lambda \mid X) = \frac{p(X \mid \lambda) p(\lambda)}{\int p(X \mid \lambda) p(\lambda) d\lambda}.$$

Step 2a. Separate the constant in the numerator of Step 2.

$$\begin{aligned} p(X \mid \lambda) p(\lambda) &= \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} = \frac{e^{-N\lambda} \lambda^S}{\prod_i x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} \\ &= \underbrace{\frac{b^a}{\Gamma(a) \prod_i x_i!}}_C \cdot \underbrace{\lambda^{a'-1} e^{-b'\lambda}}_{g(\lambda)}. \end{aligned}$$

Step 2b. Find the known density for g from Step 2a.

$$\mathcal{G}(\lambda \mid a', b') = \frac{b'^{a'}}{\Gamma(a')} \lambda^{a'-1} e^{-b'\lambda} \implies g(\lambda) = \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b').$$

Step 2c. Substitute Steps 2a and 2b into Step 2.

$$p(\lambda \mid X) = \frac{C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b')}{C \frac{\Gamma(a')}{b'^{a'}} \int \mathcal{G}(\lambda \mid a', b') d\lambda} = \frac{C \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b')}{C \frac{\Gamma(a')}{b'^{a'}}} = \mathcal{G}(\lambda \mid a + S, b + N).$$

Step 3. Substitute Step 2c into Step 1.

$$\mathbb{E}[\lambda \mid X] = \int \lambda \mathcal{G}(\lambda \mid a', b') d\lambda = \frac{b'^{a'}}{\Gamma(a')} \int \lambda^{a'} e^{-b'\lambda} d\lambda.$$

Step 3a. Find the integral for Step 3 with the same method.

$$\lambda^{a'} e^{-b'\lambda} = \frac{\Gamma(a' + 1)}{b'^{a'+1}} \mathcal{G}(\lambda \mid a' + 1, b') \implies \int \lambda^{a'} e^{-b'\lambda} d\lambda = \frac{\Gamma(a' + 1)}{b'^{a'+1}}.$$

Step 3b. Substitute Step 3a into Step 3.

$$\begin{aligned} \mathbb{E}[\lambda \mid X] &= \frac{b'^{a'}}{\Gamma(a')} \cdot \frac{\Gamma(a' + 1)}{b'^{a'+1}} = \frac{a' \Gamma(a')}{\Gamma(a') b'} & (\Gamma(a' + 1) = a' \Gamma(a')) \\ &= \frac{a'}{b'} = \frac{a + S}{b + N}. \end{aligned}$$

$$\hat{\lambda}_B = \frac{a + S}{b + N}.$$

Conclusion.

- *Conjugate prior:* gamma prior $\times$ Poisson likelihood $\rightarrow$ gamma posterior. The update only changes the parameters: $a \rightarrow a + S$ (events), $b \rightarrow b + N$ (observations).
- The estimate is a weighted mean of the prior mean and the estimate $\bar{x}$ from Problem 4:

$$\frac{a + S}{b + N} = \frac{a}{b + N} + \frac{S}{b + N} = \frac{b}{b + N} \cdot \frac{a}{b} + \frac{N}{b + N} \cdot \bar{x} \xrightarrow{N \rightarrow \infty} \bar{x}.$$

- If $a, b \rightarrow 0$, then $p(\lambda) \propto \lambda^{a-1} e^{-b\lambda} \rightarrow 1/\lambda$, the non-informative prior from the notes below, and $\hat{\lambda}_B \rightarrow S/N = \bar{x}$.

Notes on comparing the estimates from Problems 4 and 5.

1. For more about non-informative priors, see Section 5.4 “Priors” in Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*.
2. The prior $p(\lambda) \propto \frac{1}{\lambda}$ is a uniform prior on $\log \lambda$. To see this, use the change-of-variables rule for the density of a function of a random variable. See, for example, [this article](#).